

Ashray Saxena

B.E. Mechanical Engineering (Hons.) · Birla Institute of Technology and Science, Pilani

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RESEARCH INTERESTS

Computational aeroelasticity and unsteady aerodynamics; high-fidelity CFD and fluid–structure interaction; structural and combustion acoustics; data-driven reduced-order modelling and uncertainty quantification for design; aircraft and propulsion-system design. My broader aim is to build fast, calibrated surrogates for expensive high-fidelity physics so that stability boundaries and design margins become interactive design objects rather than post-hoc checks.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani — B.E. Mechanical Engineering 2023 – 2027 (expected)

- Cumulative GPA: 3.51 / 4.00. Coursework spanning solid mechanics, thermofluids, propulsion, control and optimization.
- **Graduate-level coursework:** Aeroelasticity; Gas Dynamics; Jet Propulsion; Guidance, Navigation & Control; Computational Fluid Dynamics; Prime Movers & Fluid Machines; Finite Element Method; Engineering Optimization; Advanced Mechanics of Solids; Heat Transfer; Robotics.

UPCOMING RESEARCH APPOINTMENTS

RWTH Aachen University — Institute of Aerospace Systems (ILR) Aug 2026 – Feb 2027

Bachelor's Thesis: Preliminary Aircraft Design of Hydrogen Aircraft Concepts (*Advisor: Univ.-Prof. Dr.-Ing. Eike Stumpf*)
Aachen, Germany

- Cabin and hydrogen-tank layout integration for hydrogen-powered transport aircraft; trade-off and optimization studies of tank/cabin configurations and their impact on aircraft performance and feasibility, using the institute's in-house preliminary-design tool UNICADO.

University of Manchester — Mechanical, Aerospace & Civil Engineering Jul 2026 – Dec 2026

Visiting Research Intern: Development of a Novel Atomiser for High-Speed Propulsion Systems (*Advisor: Dr. Kuppuraj Rajamanickam*) Remote

- Design of a new atomiser concept for high-speed propulsion, coupling spray/atomisation physics to injector geometry for improved mixing and combustion stability.

SELECTED RESEARCH PROJECTS

Uncertainty-Calibrated Prediction of the Transonic Flutter Boundary via an Actively-Sampled ROM Jan 2026 – present

Independent framework; targeting the IEEE Aerospace Conference first author

- **Objective.** Predict the transonic flutter boundary directly (flutter speed index and damping margin) as a calibrated function of flight/design parameters, replacing hundreds of hour-long coupled solves with sub-millisecond queries.
- **Methods & tools.** p - k aeroelastic solver with Theodorsen unsteady aerodynamics (subsonic baseline); high-fidelity coupled SU2 fluid–structure-interaction with a Newmark structural integrator and a custom settable-elastic-axis 2-DOF coupler (Python, MPI); Gaussian-process operator on a tanh-squashed growth-rate field; level-set ($\sigma = 0$) boundary extraction; split-conformal uncertainty calibration; variance/boundary-distance active-learning acquisition.
- **Datasets.** AGARD 445.6 weakened-wing experimental flutter data (Yates, 1988) for subsonic validation; a self-generated, actively-sampled SU2 FSI snapshot database (growth rate vs. Mach and elastic-axis location), resumable via a durable cache.
- **Key assumptions.** Modal (typical-section) structural idealisation; matched-frequency p - k ; growth-rate *sign* (not magnitude) is the trusted label far from the boundary; squared-exponential GP kernel; exchangeability for conformal coverage.
- **Outcomes & contributions.** Linear baseline reproduces the subsonic AGARD flutter speed index to within $\sim 12\%$; 2-D SU2 runs recover a transonic boundary and the Isogai Case-A deep-dip-plus-restabilisation; the tanh transform cuts the few-sample boundary-fit error by an order of magnitude; conformal band attains target coverage; active learning reaches a converged boundary in materially fewer solves than random sampling. Open, reproducible pipeline (88 passing tests). Honest status: 3-D AGARD path runs but is gated on a mesh-independence study.
- **Key references.** Theodorsen (NACA TR-496, 1935); Yates (AGARD R-765, 1988); Isogai (AIAA J., 1979); Economon et al., SU2 (AIAA J., 2016); Rasmussen & Williams (2006); Vovk et al. (2005).

- **Objective.** Extend the asymptotic coupled-wavenumber theory of a fluid-loaded flexible plate to a duct bounded by two flexible plates, and add a mean flow.
- **Methods & tools.** Closed-form coupled dispersion relation with exact symmetric/antisymmetric factorisation; first-order asymptotic expansion in the fluid-loading parameter $\varepsilon = \rho a/m$ with local $\sqrt{\varepsilon}$ windows at veering/cut-on; Pridmore–Brown equation with plate-admittance (Robin) boundary conditions; Fredholm solvability for the $O(M)$ correction and an $O(M^2)$ convection/shear split; inviscid (Frobenius) and viscous critical-layer analysis. Symbolic verification (SymPy) plus shooting and spectral root-finders.
- **Datasets.** None external; all analytical results validated against exact numerical roots of the dispersion relation via three independent numerical methods.
- **Key assumptions.** Thin elastic plates; 2-D, time-harmonic, linear acoustics; low Mach number; Poiseuille mean-flow profile $f(\eta) = 4\eta(1 - \eta)$; small- ε ordering.
- **Outcomes & contributions.** A closed-form $O(M)$ correction to every coupled branch; the critical-layer threshold $M_{\text{crit}} = \Omega/\xi$; the result that discrete wall modes are *not* absorbed by the critical layer (Stokes-controlled $Re^{-1/2}$ decay) and that the system is convectively stable. Three numerical methods agree to ~ 1 part in 10^9 . Framed honestly as an incremental, mean-flow contribution over the established quiescent catalogue.
- **Key references.** Pridmore-Brown (JFM, 1958); Sarkar & Sonti (JSV, 2007); Prakash & Sonti (JSV, 2016); Brambley (JSV, 2009); Fahy & Gardonio (2007).

Continuous-Wave Rotor Pressure Exchanger for Supercritical CO₂ Cycles

Jan 2026 – present

Under review, *International Journal of Refrigeration*

author

- **Objective.** Improve direct-contact pressure exchange for sCO₂ thermal/refrigeration cycles by managing transient pressure waves in the rotor channels.
- **Methods & tools.** Transient pressure-wave analysis of rotor-channel exchange; comparison against conventional rotary-blade architectures; mixing-loss and thermal-management modelling.
- **Datasets.** Supercritical-CO₂ thermophysical property data and cycle operating conditions.
- **Key assumptions.** Continuous-wave (quasi-periodic) exchange; idealised channel wave dynamics; representative sCO₂ cycle states.
- **Outcomes & contributions.** A continuous-wave exchange mechanism achieving a **49% increase in isentropic efficiency** over conventional rotary-blade designs, with reduced mixing losses and improved thermal management.

Frequency-Domain Analysis of Thermoacoustic Coupling in Liquid-Propellant Rocket Engines

May 2025 – Nov 2025

IEEE Aerospace Conference 2026 (grew out of the DHIO internship)

author

- **Objective.** Identify chamber geometry and damping that suppress thermoacoustic (combustion) instability during the ignition transient.
- **Methods & tools.** High-fidelity pressure-acoustics, frequency-domain Helmholtz model on an axisymmetric chamber/nozzle (COMSOL Multiphysics); frequency sweep over the ignition band; Rayleigh-criterion-based mitigation by spatially decoupling pressure and heat-release fluctuations; mesh-free CFD for the underlying flow.
- **Datasets.** Combustion-state mean-flow properties (density, sound speed); a 500–3000 Hz forcing sweep; resolved near-wall mesh.
- **Key assumptions.** Linear acoustics; axisymmetry; harmonic forcing; mean state frozen over an acoustic cycle.
- **Outcomes & contributions.** A fast frequency-domain screen that maps the total sound-pressure-level field, locates susceptible antinodes relative to the heat-release zone, and selects geometry/damping that pushes the Rayleigh integral negative, ahead of any fully coupled reacting-flow simulation.
- **Key references.** Rayleigh (1878); Culick (AGARD, 2006); Harrje & Reardon (NASA SP-194, 1972).

Rapid Aerodynamic Optimization of Engine Nacelles (CST + Viscous–Potential Solver)

2025

Technical paper

first author

- **Objective.** Make nacelle lip/cowl drag optimization interactive by replacing RANS sweeps with a fast, compressibility-aware screen.
- **Methods & tools.** Class Shape Transformation (Kulfan) parameterisation of the lip and cowl with Bernstein shape functions; an in-house viscous–potential solver (finite-difference stream function for local Mach/pressure, boundary-layer Reynolds-number model for empirical skin friction); separate form/friction drag accounting; automated lip-morphing optimizer. Implemented as an integrated Python framework.
- **Datasets.** Self-generated viscous–potential and finite-difference flow solutions across lip-flare-angle sweeps; published supercritical/multi-element-airfoil test data as reference context.

- **Key assumptions.** 2-D fluid domain; potential core plus boundary-layer correction; continuity-based capture ($\dot{m} = \rho AV$); localised lip morphing before any Navier–Stokes refinement.
- **Outcomes & contributions.** A design filter that localises drag minima and incipient separation, delivering significant form-drag reductions at a fraction of RANS cost; resolves the non-monotonic drag-vs-flare trade that brute-force optimizers mis-map.
- **Key references.** Kulfan (J. Aircraft, 2008); Seddon & Goldsmith, *Intake Aerodynamics* (1999); Drela, XFOIL (1989).

Conjugate Thermal Management of a Liquid-Cooled 14S4P Battery Pack

Oct 2025 – Dec 2025

COMSOL Multiphysics study (Pininfarina Battista pack); ISHRAE

author

- **Objective.** Verify that a plate-and-pipe liquid-cooling design holds cell-to-cell temperature uniformity in a high-power automotive pack under continuous discharge.
- **Methods & tools.** Coupled electrochemical–thermal model (COMSOL Battery Design + Heat Transfer): ohmic/activation/concentration losses as volumetric heat sources in the energy equation; non-isothermal turbulent flow coupled to a non-isothermal pipe-flow cooling network; temperature-dependent Arrhenius state-of-health (ageing) model.
- **Datasets.** Pininfarina Battista pack public manufacturer data (~ 120 kWh, ~ 630 V, $174s40p \approx 6960$ cells, Li-Mn-Ni chemistry); a 14S4P simulated subset ($8 \text{ modules} \times 56 \text{ cells}$); transient discharge cycle.
- **Key assumptions.** Lumped per-cell electrochemistry; representative cycle; Arrhenius-controlled degradation; conjugate heat transfer between cells, plate and coolant.
- **Outcomes & contributions.** Temperature rise held below $\sim 8^\circ\text{C}$ with no hotspots (peak $\approx 27.6^\circ\text{C}$); voltage and state of charge uniform across modules; state of health kept above 0.80, decay correlating with local temperature as predicted. A reusable design tool for sizing pipe layout, flow rate and cell spacing against a target duty cycle.

PROFESSIONAL RESEARCH EXPERIENCE

LAT Aerospace — Aeromodelling & Avionics Intern; Gas Turbine Engine Team

Dec 2025 – present

- **SkyNET high-lift airfoil framework.** Inverse-design and optimization framework that synthesises high-lift airfoils for blown-wing and high-aspect-ratio UAS from target lift/drag coefficients, evaluated with viscous–inviscid panel methods (XFOIL family) and SU2; designed single-slotted fowler-flap/Coanda devices with their deployment kinematics, using the slot-height-to-curvature ratio h/R to keep the blown jet attached.
- **FADEC & engine simulation.** Developing a modular Full-Authority Digital Engine Controller for turbojets on an STM32 Nucleo-H743ZI2, with a C++ software-in-the-loop turbojet simulator for controller development; contributed to a hybrid UAV powertrain pairing microturbojets with electric ducted fans on a 130 kg aircraft.

DHIO Research & Engineering Pvt. Ltd. — CFD Engineering Intern

May 2025 – Jul 2025

- High-fidelity mesh-free CFD of liquid-propellant rocket-engine flows; investigated thermoacoustic instability and combustion dynamics by frequency-domain analysis (seed of the IEEE AeroConf work) and evaluated chamber acoustic coupling for damping/ignition stability.
- Built a transient, arbitrary-Lagrangian–Eulerian moving-mesh CFD study of a drag-reduction system, quantifying the load change between closed and open states of an actuated upper element.

India Space Lab — Satellite Communication Intern

Jul 2025 – Aug 2025

- Engineered the communication subsystem for CubeSat-class nanosatellites (link-budget optimization, signal integrity); developed and deployed a Telemetry, Tracking & Command (TT&C) system; supported transponder/antenna-array integration to orbital communication standards.

ACADEMIC PROJECTS

Aerodynamic Characterization of STOL Configurations — IIT Bombay

Nov 2025 – Jan 2026

- Evaluated counter-circulation and distributed-integrated-propulsion blown-wing architectures; generated C_L – α polars via parametric studies, quantifying stall delay, boundary-layer-control lift enhancement, and propeller-slipstream/wing-chord interaction in low-speed flight.

Czerny–Turner Raman Spectrophotometer — DST Nidhi Prayas Labs

May 2024 – Nov 2024

- Architected a high-precision Raman spectrophotometer (Czerny–Turner monochromator, laser optics) for non-destructive detection of organic biosignatures; optimised optical throughput and signal-to-noise; integrated diffraction gratings and a CCD for high-resolution acquisition.

Design & Structural Analysis of a Modular ATV Monocoque — ASME Student Chapter (BAJA SAE) Sep 2024 – Feb 2025

- Engineered a high-torsional-stiffness monocoque chassis (mass-optimised to BAJA SAE standards) with anti-Ackermann steering geometry; iterative kinematic/stress analysis of an integrated high-efficiency geartrain under dynamic loading.

5-DOF Manipulator with Reconfigurable End-Effectors — IEEE RASSE 2025

Sep 2023 – Feb 2024

- Designed and prototyped a five-DOF arm with a hot-swappable end-effector interface and a rotational-base control architecture; analysed workspace envelope and kinematic constraints for tasks from microscopic sampling to warehouse automation.

TECHNICAL LEADERSHIP

Team Captain — CRISS Robotics, Space Engineering & Robotics Team, BITS Pilani

Jan 2024 – present

- Directs a 50+ engineer team across four subsystems and a \$20,000+ budget; led the first Indian delegation to qualify for the Australian Rover Challenge (University of Adelaide).
- Podium at the European Rover Challenge 2025 and 3rd place at the International Rover Design Challenge 2025; "Best Project Presentation" at APOGEE 2025 for multispectral planetary-exploration drones.
- Integrated ROS with OpenCV for an autonomous, camera-calibrated spectroscope; designed widefield-imaging and IR-spectrometer payloads and the rover sample-collection system for simulated Martian terrain.

ENTREPRENEURSHIP

Chief Technical Officer — AtmosClear (Smart Autonomous Air Purification)

Dec 2024 – Feb 2026

- Designed and built a working smart autonomous mobile air purifier (household/industrial); under review for a design patent; funded under the DST Nidhi Prayas programme, Government of India.

Chief Technical Officer — Space-Mined (In-Situ Resource Utilisation)

Jun 2025 – Oct 2025

- Space-mission design for laser-induced thermal processing of lunar regolith (in-situ metal/oxygen extraction); developed a semi-scale TRL-3 prototype with a Luxembourg startup accelerator.

TEACHING & MENTORING

Undergraduate Teaching Assistant — Dept. of Mechanical Engineering, BITS Pilani

Aug 2025 – present

- Instruction support for 150+ students in Mechanics of Solids (ME F211), Gas Dynamics (ME F434) and Quality Control, Assurance & Reliability; tutorials/office hours in high-speed aerodynamics and structural mechanics; coursework evaluation.

Research Writing Coach — RISE Global Education

Sep 2025 – Jan 2026

- Mentored 10+ secondary students through the scholarly-publishing lifecycle (literature review, hypothesis formulation, manuscript drafting) with workshops on academic integrity and citation.

PUBLICATIONS & MANUSCRIPTS

- A. Saxena, "Frequency-Domain Analysis of Thermoacoustic Coupling in Liquid-Propellant Rocket Engines," *IEEE Aerospace Conference 2026* (accepted/forthcoming).
- A. Saxena, "Continuous-Wave Rotor Pressure Exchanger for Supercritical CO₂ Cycles," *International Journal of Refrigeration* (under review).
- A. Saxena, "Coupled Dispersion of a Two-Flexible-Plate Structural-Acoustic Waveguide with Low-Mach Poiseuille Flow" (manuscript in preparation).
- A. Saxena, "Rapid Aerodynamic Optimization of Engine Nacelles" (technical paper).
- Co-author, "5-DOF Manipulator with Reconfigurable End-Effectors," *IEEE RASSE 2025*.

TECHNICAL SKILLS

CFD / simulation: SU2, ANSYS, OpenFOAM, COMSOL Multiphysics, Star-CCM+, SimScale, ParaView, ultraFluidX, Simulink.

Aerodynamics / design: XFOIL, XFLR5, OpenVSP, AeroSandbox, OpenAeroStruct, NeuralFoil, pymead; CST/Bernstein parameterisation; UNICADO (incoming).

CAD: SolidWorks, Fusion 360, Siemens NX, CATIA V5, PTC Creo, Onshape, AutoCAD.

Programming / methods: Python, C, C++, MATLAB; Gaussian processes, conformal prediction, NSGA-II, active learning, POD/DMD; Git.

Embedded / robotics: ROS2, STM32, KiCAD, Altium, YOLOv8, PointNet++.

HONORS & SCHOLASTIC ACHIEVEMENTS

- **NTSE Scholar (AIR 4):** National Talent Search Examination, among 350,000+ candidates; centum in Mental Ability.
- **KVPY Scholar (AIR 189):** Kishore Vaigyanik Protsahan Yojana, among 200,000+; centum in Mathematics & Physics.

- **Astronomy Olympiad (IOQA), AIR 29:** shortlisted for the HBCSE training camp (2023).
- **Mathematics Olympiad (IOQM):** AIR Top 100 for three consecutive years.

References available upon request.